WORKED SOLUTION

STEP I 2016, Question 6

Pure • Vector Geometry

STEP I 2016, Question 6

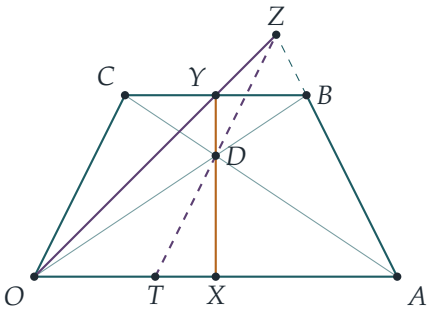
Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s1-q6

The key idea

Every point in the figure is a meeting point of two lines, so we write each line in vector form and equate. Because **a** and **b** are not parallel, any vector can be written as $\alpha\mathbf{a} + \beta\mathbf{b}$ in only one way, so equating two expressions for the same point lets us compare coefficients of **a** and **b** separately. Working in the basis **a**, **b** throughout keeps every step to two short linear equations.

Solution



The forms of c and x

Since CB is parallel to OA, the vector $\overrightarrow{BC} = \mathbf{c} - \mathbf{b}$ is a scalar multiple of **a**, say $\mathbf{c} - \mathbf{b} = k\mathbf{a}$, which gives $\mathbf{c} = k\mathbf{a} + \mathbf{b}$. Going round the quadrilateral $O \to A \to B \to C$, the side BC must run in the opposite direction to OA (if it ran the same way, the sides AB and CO would cross and $OABC$ would not be a genuine quadrilateral), and $C \neq B$. So

$$k < 0.$$

Any negative value is possible; $k = -1$ is the parallelogram.

Since X lies on OA, **x** is a multiple of **a**, say $\mathbf{x} = m\mathbf{a}$, and X strictly between O and A

means

$$0 < m < 1.$$

Also $\mathbf{a}$ and $\mathbf{b}$ are not parallel, as O, A, B are not collinear.

The point Z

Finding D. Points of OB are $\lambda\mathbf{b}$; points of AC are

$$\mathbf{a} + \mu(\mathbf{c} - \mathbf{a}) = (1 - \mu + \mu k)\mathbf{a} + \mu\mathbf{b}.$$

Comparing coefficients, $\lambda = \mu$ and $1 - \mu(1 - k) = 0$. As $k < 0$, $1 - k > 0$, so $\mu = \frac{1}{1-k}$ and

$$\mathbf{d} = \frac{\mathbf{b}}{1-k}.$$

Finding Y. Points of BC are $\mathbf{b} + t\mathbf{a}$. Points of XD are

$$m\mathbf{a} + s(\mathbf{d} - m\mathbf{a}) = m(1-s)\mathbf{a} + \frac{s}{1-k}\,\mathbf{b}.$$

The $\mathbf{b}$-coefficients give $s = 1 - k$, and then the $\mathbf{a}$-coefficient is $m(1 - (1-k)) = mk$. So

$$\mathbf{y} = \mathbf{b} + mk\,\mathbf{a}.$$

Finding Z. Points of OY are $\rho(\mathbf{b} + mk\mathbf{a})$; points of AB are $\mathbf{a} + \sigma(\mathbf{b} - \mathbf{a}) = (1-\sigma)\mathbf{a} + \sigma\mathbf{b}$. Comparing coefficients, $\rho = \sigma$ and $\rho mk = 1 - \sigma$, so $\rho(mk + 1) = 1$. Provided $mk + 1 \neq 0$,

$$\mathbf{z} = \frac{\mathbf{b} + mk\,\mathbf{a}}{mk+1},$$

as required. (If $mk = -1$, which can happen when $k < -1$, then $\mathbf{y} = \mathbf{b} - \mathbf{a}$, so OY is parallel to AB and Z does not exist.)

Ask yourself Is Z on the side AB itself?

No: $\sigma = \frac{1}{mk+1} > 1$ because $-1 < mk < 0$, so Z lies on AB extended beyond B, as in the figure. That doesn't matter, since the question concerns the lines.

The point T and the two relations

Points of DZ are $\mathbf{d} + \tau(\mathbf{z} - \mathbf{d})$. The $\mathbf{b}$-coefficient is

$$\frac{1}{1-k} + \tau\left(\frac{1}{mk+1} - \frac{1}{1-k}\right) = \frac{1}{1-k} - \tau\,\frac{k(1+m)}{(mk+1)(1-k)},$$

using $(1-k) - (mk+1) = -k(1+m)$. For T on OA this must vanish, so $\tau = \dfrac{mk+1}{k(1+m)}$ (recall $k \neq 0$). The $\mathbf{a}$-coefficient is then

$$\tau \cdot \frac{mk}{mk+1} = \frac{mk+1}{k(1+m)} \cdot \frac{mk}{mk+1} = \frac{m}{1+m}.$$

Hence

$$\mathbf{t} = \frac{m}{1+m}\,\mathbf{a}.$$

Remarkably, k has disappeared: T depends only on where X is.

Now write $a = |\mathbf{a}| = OA$. Since $0 < \frac{m}{1+m} < 1$, T lies between O and A, so

$$OT = \frac{ma}{1+m}, \qquad TA = a - \frac{ma}{1+m} = \frac{a}{1+m}, \qquad OX = ma.$$

Therefore

$$OT \times OA = \frac{ma^2}{1+m} = ma \times \frac{a}{1+m} = OX \times TA,$$

and

$$\frac{1}{OX} + \frac{1}{OA} = \frac{1}{ma} + \frac{1}{a} = \frac{1+m}{ma} = \frac{1}{OT},$$

which are both results required.

Checks

Try the trapezium in the figure: $\mathbf{a} = (4,0)$, $\mathbf{b} = (3,2)$, $k = -\frac{1}{2}$, $m = \frac{1}{2}$. Then $D = (2, \frac{4}{3})$, $Y = (2,2)$, $Z = (\frac{8}{3}, \frac{8}{3})$ and $T = (\frac{4}{3}, 0)$; you can confirm that T, D, Z are collinear, and $\frac{1}{OT} = \frac{3}{4} = \frac{1}{2} + \frac{1}{4} = \frac{1}{OX} + \frac{1}{OA}$. A computer check on 20,000 random trapezia agreed with every formula above to within 10^{-10}. The limiting case $m \to 1$ ($X \to A$) gives $OT = \frac{1}{2}OA$, consistent with $\frac{1}{OT} = \frac{2}{OA}$.

Common slips

- Taking $k > 0$: with $\mathbf{c} = \mathbf{b} + k\mathbf{a}$ and $k > 0$ the "quadrilateral" crosses itself.
- Equating coefficients without saying why it is allowed: it relies on $\mathbf{a}$ and $\mathbf{b}$ not being parallel.
- Using the same parameter letter for two different lines, which forces a false coincidence.

Beyond STEP

The fact that T does not depend on k (nor on the angle between $\mathbf{a}$ and $\mathbf{b}$) shows this is really a theorem of affine, and indeed projective, geometry: only incidences and ratios along a line matter. The relation $\frac{1}{OT} = \frac{1}{OX} + \frac{1}{OA}$ is the sort of statement that cross-ratios and harmonic ranges handle elegantly, and the same construction, repeated, places points at $\frac{m}{1+2m}\mathbf{a}$, $\frac{m}{1+3m}\mathbf{a}$, and so on. For more, see Coxeter and Greitzer, *Geometry Revisited*, or Coxeter, *Projective Geometry*.

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